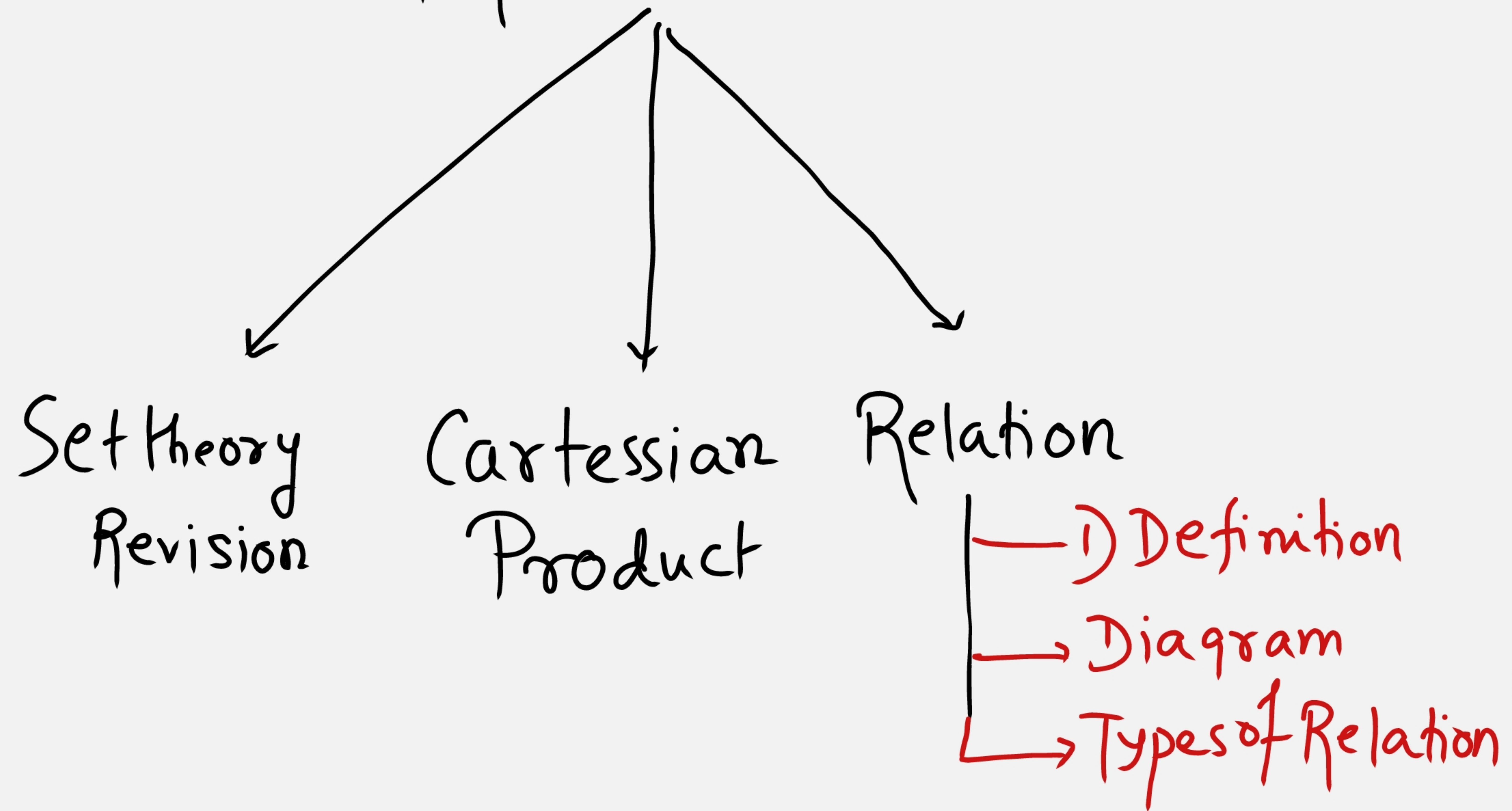
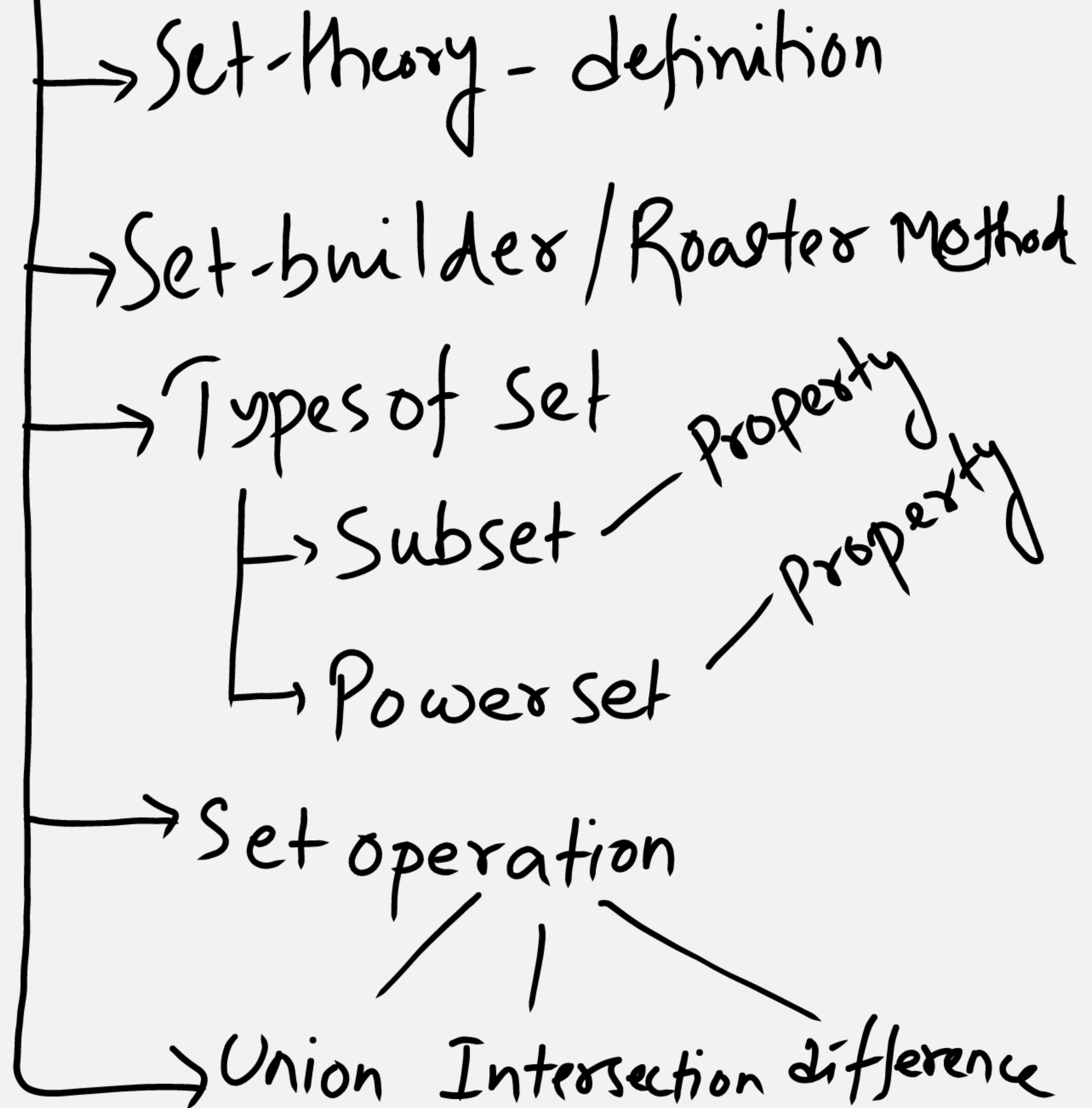


Set-Theory & Relation

Topics to be covered.



Recap- Previous Lecture



Cartesian product

$$A = \{a, b, c\}, B = \{1, 2, 3\}$$

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3), (c, 1), (c, 2), (c, 3)\}$$

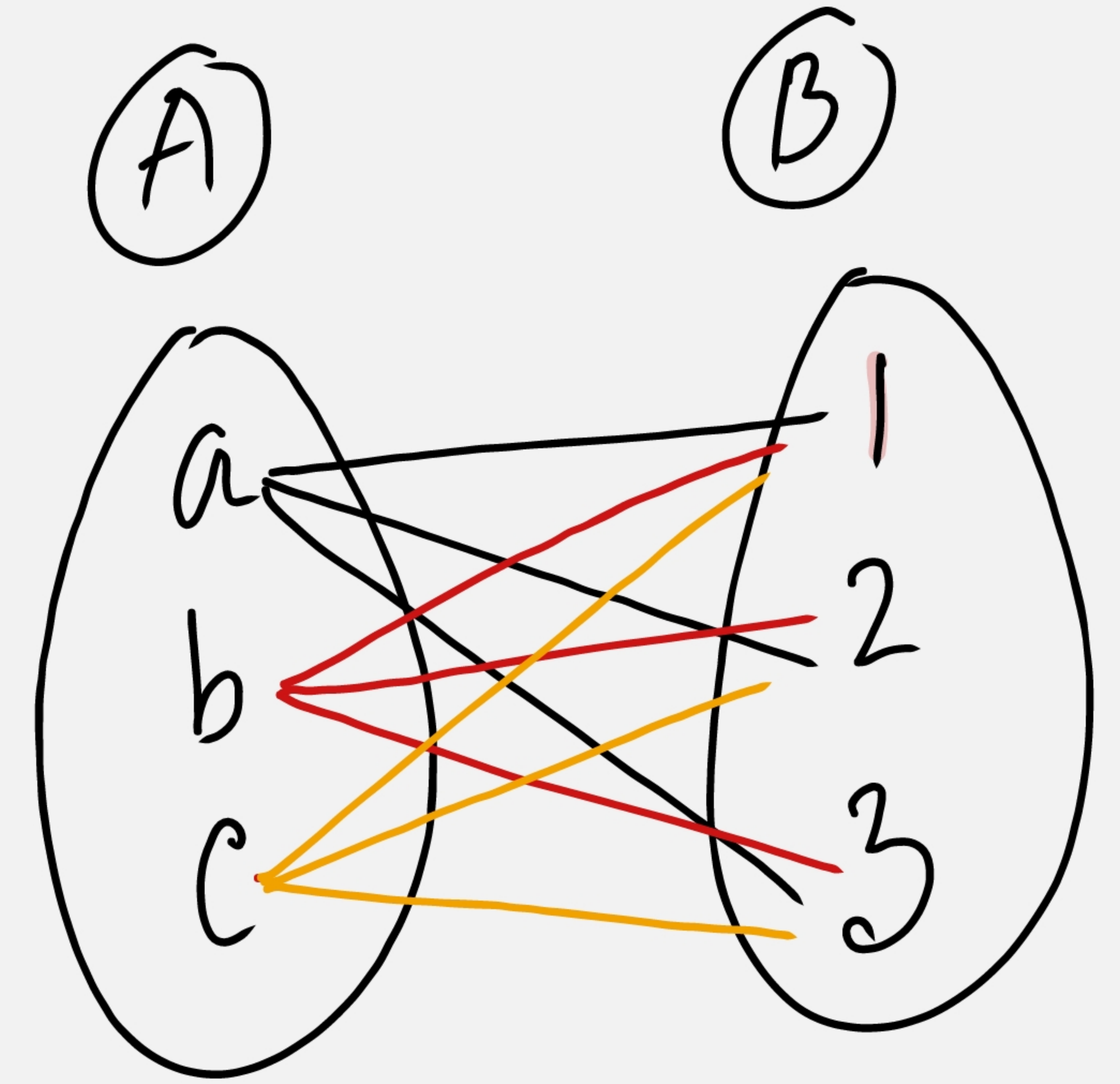
Properties:

$$B \times A = \{(1, a), (1, b), (1, c), \dots\}$$

If set A have m-element and B-have n element

Then total no. of Cartesian Product is $= m \times n$

$$|A \times B| = m \times n$$



$$A \times B = \{ \quad \}$$
$$R = \{(a, 1), (b, 1), (c, 1)\}$$

Relation

Let, A & B are two sets Then every relation from A to B is a Subset of $A \times B$.

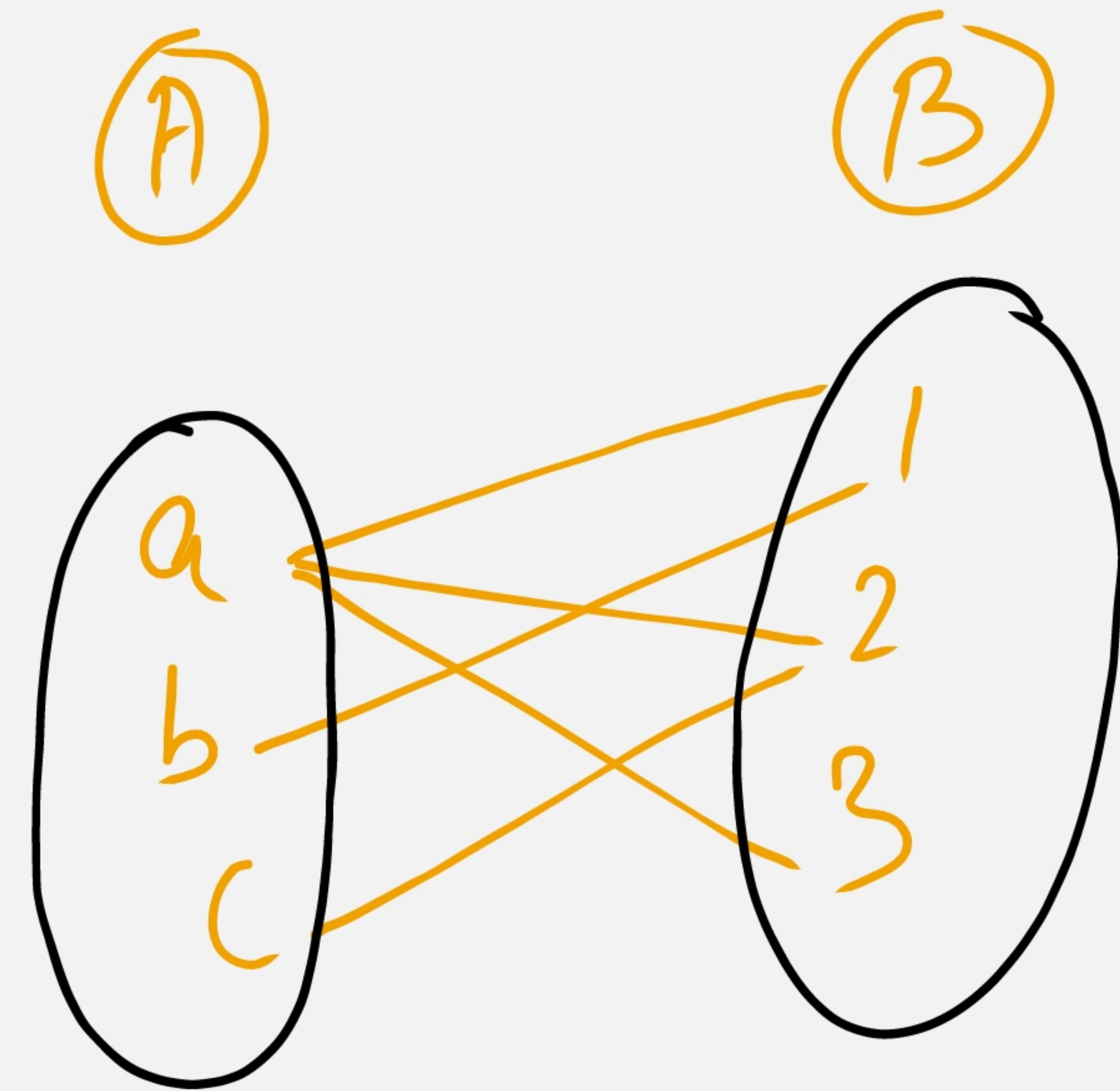
No. of Relation = $2^{m \times n}$

Subset = (2^n)

* $A \times B \neq B \times A$

if set are equal ($A = B$) then $A \times B = B \times A$ -

if set are null then $A \times B = B \times A$ -



Types of Relation

i) Reflexive Relation

ii) Symmetric Relation

iii) Irreflexive Relation

iv) Asymmetric Relation

v) Transitive Relation

vi) Equivalence Relation

Reflexive Relation

A Relation 'R' on a set 'A' is said to be Reflexive if

$$(x, x) \in R \quad \forall x \in A$$

$A = \{1, 2, 3\}$

$A \times A = \left\{ \begin{array}{l} (1,1), (1,2), (1,3) \\ (2,1), (2,2), (2,3) \\ (3,1), (3,2), (3,3) \end{array} \right\}$

(x, x) ✓

$$A \times A = m \times n = 9 \text{ element}$$

$$R_1 = \{(1,1), (2,2), (3,3)\} \quad \checkmark \quad (R)$$

$$R_2 = \{(1,1), (2,2)\} \quad \text{not Reflexive bcz} \quad \forall x \notin A \quad \checkmark$$

$$R_3 = \{(\underline{1},1), (\underline{2},2), (\underline{3},3), (1,2)\} \quad \checkmark \quad (R)$$

↓

examples ∴

$$R_1 = \{ (x, y) \mid x - y \text{ is an integer} \}$$

$$R_2 = \{ (x, y) \mid x - y \text{ is an odd no.} \}$$

check which of the above is Reflexive
on a set of all natural numbers?

R_2 is not Reflexive ✓

$$(2, 2) = 2 - 2 = 0 = \text{even number} ✓$$

(1, 1)

(2, 2)

(3, 3)

Let, $2, 3 = 2 - 3 = (-1) \times$

$(2, 2) = 2 - 2 = 0 \checkmark$ (integer)

Above R_1 is a Reflexive Relation.

How many Reflexive relation is possible?

$A = \{1, 2\}$ then $A \times A = \left\{ \begin{array}{cc} (1,1) & (1,2) \\ (2,1) & (2,2) \end{array} \right\}$

Annotations: $\rightarrow 2^2 = 4$ (pointing to the set), $\textcircled{1}$ (pointing to $(1,1)$), $\textcircled{2}$ (pointing to $(2,2)$).

$A = \{1, 2, 3\}$ then $A \times A = \left\{ \begin{array}{ccc} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3) \end{array} \right\}$

Annotations: $\rightarrow 2^6 = 64$ (pointing to the set), $\textcircled{2}$ (pointing to $(1,1)$), $\textcircled{1}$ (pointing to $(2,2)$).

$A = \{1, 2, 3, 4\}$ then $A \times A = \left\{ \begin{array}{cccc} (1,1) & (1,2) & (1,3) & (1,4) \\ (2,1) & (2,2) & (2,3) & (2,4) \\ (3,1) & (3,2) & (3,3) & (3,4) \\ (4,1) & (4,2) & (4,3) & (4,4) \end{array} \right\}$

Annotations: $\rightarrow 2^{12} = 4096$ (pointing to the set), $2, 6, 12 \rightarrow \text{Nondiagonal}$, $n^2 - n$ (circled), $\text{Total no. of Reflexive Relation} = 2$ (circled).

$R = \{ (1,1) (2,2) \}$

$\{ (1,1), (2,2) (1,2) \}$

$\{ (1,1) (2,2) (2,1) \}$

$\{ (1,1), (2,2), (1,2) (2,1) \}$

Total no. of permutation
 $= 1 \times 2 \times 2 \times 1 = 4$
 $= 2^2$

$A = n$

$A \times A = n^2$

Diagonal = n

ND = $n^2 - n$

Irreflexive Relation!!

$$\begin{aligned} \text{Total} &= R - \text{Reflexive} \\ \text{Non-reflexive} &= 2^{n^2} - 2^{n-1} \end{aligned}$$

A Relation R on a set ' A ' is said to be 'irreflexive' if $x \not R x$
 $\{x \text{ is not related to } x\} \forall x \in A$

$$\text{i.e. } (x, x) \notin R$$

$$A = \{1, 2, 3\}$$

$$\underbrace{A \times A}_{(R)} = \left\{ \begin{array}{l} (1,1) (1,2) (1,3) \\ (2,1) (2,2) (2,3) \\ (3,1) (3,2) (3,3) \end{array} \right\}$$

$$R = \{(\underline{1}, \underline{1}), (\underline{1}, \underline{2})\} \times$$

$$R = \{(\underline{1}, \underline{2}), (\underline{1}, \underline{3}), (\underline{2}, \underline{1})\} \checkmark$$

$$\left. \begin{array}{l} \text{Smallest possible} = n \\ \text{Largest } n = n^2 \end{array} \right\}$$

$$R = \{ (1,2), (1,3), (2,1), (2,3), (3,1), (3,2) \}$$

Largest element of R

$$\text{non-diagonal} = n^2 - n$$

Total no. of Irreflexive Relation =

$$2^{n^2 - n}$$